

Customer Behavior Analysis

Project Overview:-

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

Dataset Summary:-

- Rows: 3,900
- Columns: 18
- Key Features:
 1. Customer demographics (Age, Gender, Location, Subscription Status)
 2. Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 3. Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

Exploratory Data Analysis Using Python:-

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `data.info()` to check structure and `data.describe()` for summary statistics.

```
[5]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	V
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	I
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	I

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

```
[11]: data.isnull().sum()
```

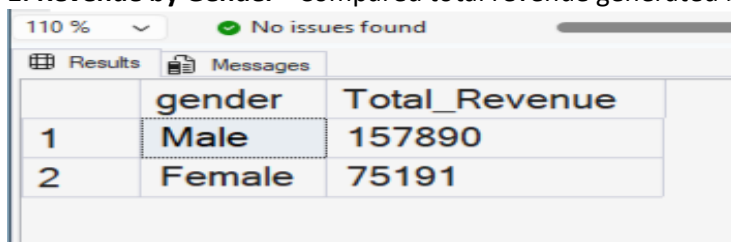
```
[11]: Customer ID      0
Age                0
Gender             0
Item Purchased     0
Category           0
Purchase Amount (USD) 0
Location           0
Size              0
Color             0
Season            0
Review Rating      37
Subscription Status 0
Shipping Type      0
Discount Applied   0
Promo Code Used    0
Previous Purchases 0
Payment Method     0
Frequency of Purchases 0
dtype: int64
```

- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 1. Created age_group column by binning customer ages.
 2. Created purchase_frequency_days column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to SQL Server and loaded the cleaned DataFrame into the database for SQL analysis.

Data Analysis using SQL (Business Transactions):-

We performed structured analysis in SQL Server to answer key business questions:

1. Revenue by Gender - Compared total revenue generated by male vs. female customers.



The screenshot shows a SQL Server query results window with a zoom level of 110%. The status bar indicates 'No issues found'. The results are displayed in a table with two columns: 'gender' and 'Total_Revenue'. There are two rows: 'Male' with a total revenue of 157890, and 'Female' with a total revenue of 75191.

	gender	Total_Revenue
1	Male	157890
2	Female	75191

2. High-Spending Discount Users - Identified customers who used discounts but still spent above the average purchase amount.



The screenshot shows a SQL Server query results window with a zoom level of 110%. The status bar indicates 'No issues found'. The results are displayed in a table with four columns: 'customer_id', 'purchased_amount', and 'discount_applied'. There are 18 rows, all showing a purchased amount of 100 and a discount applied of 'Yes'.

	customer_id	purchased_amount	discount_applied
10	616	100	Yes
11	770	100	Yes
12	862	100	Yes
13	1209	100	Yes
14	1301	100	Yes
15	1406	100	Yes
16	1413	100	Yes
17	1422	100	Yes
18	1457	100	Yes

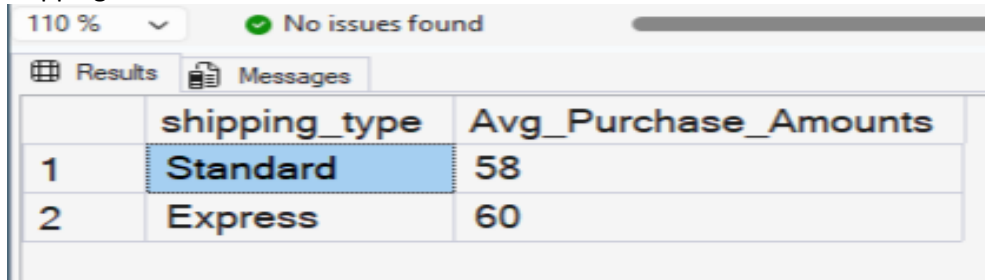
3. Top 5 Products by Rating - Found products with the highest average review ratings.



The screenshot shows a SQL Server query results window with a zoom level of 110%. The status bar indicates 'No issues found'. The results are displayed in a table with two columns: 'item_purchased' and 'Avg_Review_Rating'. There are 5 rows, with 'Gloves' having the highest average rating of 3.86.

	item_purchased	Avg_Review_Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Handbag	3.78

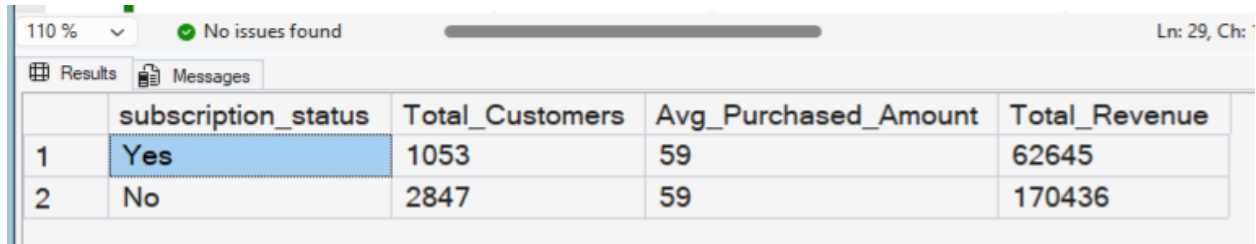
4. Shipping Type Comparison - Compared average purchase amounts between Standard and Express shipping.



The screenshot shows a data table with a zoom level of 110% and a status of 'No issues found'. The table has two columns: 'shipping_type' and 'Avg_Purchase_Amounts'. There are two rows: 'Standard' with an average purchase amount of 58, and 'Express' with an average purchase amount of 60. The 'Standard' row is highlighted in blue.

	shipping_type	Avg_Purchase_Amounts
1	Standard	58
2	Express	60

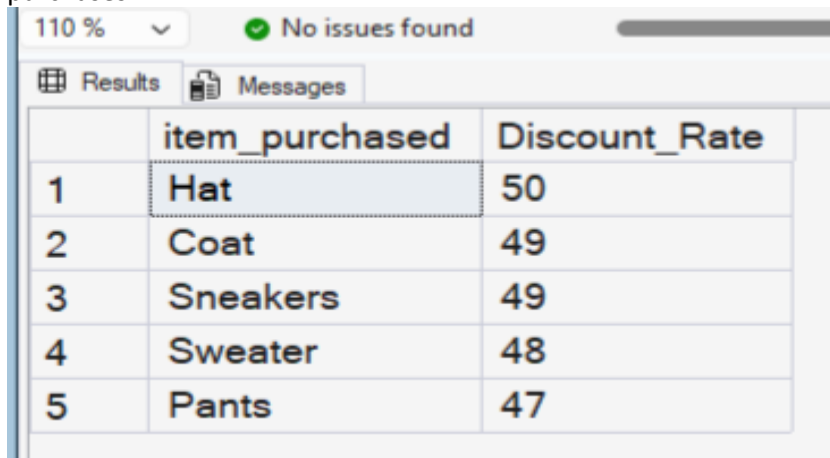
5. Subscribers vs. Non-Subscribers - Compared average spend and total revenue across subscription status.



The screenshot shows a data table with a zoom level of 110% and a status of 'No issues found'. The table has four columns: 'subscription_status', 'Total_Customers', 'Avg_Purchased_Amount', and 'Total_Revenue'. There are two rows: 'Yes' with 1053 total customers, an average purchase amount of 59, and a total revenue of 62645; and 'No' with 2847 total customers, an average purchase amount of 59, and a total revenue of 170436. The 'Yes' row is highlighted in blue.

	subscription_status	Total_Customers	Avg_Purchased_Amount	Total_Revenue
1	Yes	1053	59	62645
2	No	2847	59	170436

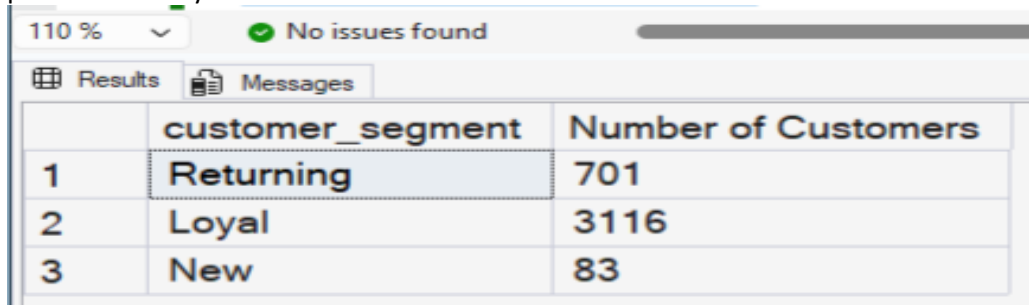
6. Discount-Dependent Products - Identified 5 products with the highest percentage of discounted purchases.



The screenshot shows a data table with a zoom level of 110% and a status of 'No issues found'. The table has two columns: 'item_purchased' and 'Discount_Rate'. There are five rows: 'Hat' (50), 'Coat' (49), 'Sneakers' (49), 'Sweater' (48), and 'Pants' (47). The 'Hat' row is highlighted in blue.

	item_purchased	Discount_Rate
1	Hat	50
2	Coat	49
3	Sneakers	49
4	Sweater	48
5	Pants	47

7. Customer Segmentation - Classified customers into New, Returning, and Loyal segments based on purchase history



The screenshot shows a data table with a zoom level of 110% and a status of 'No issues found'. The table has two columns: 'customer_segment' and 'Number of Customers'. There are three rows: 'Returning' (701), 'Loyal' (3116), and 'New' (83). The 'Returning' row is highlighted in blue.

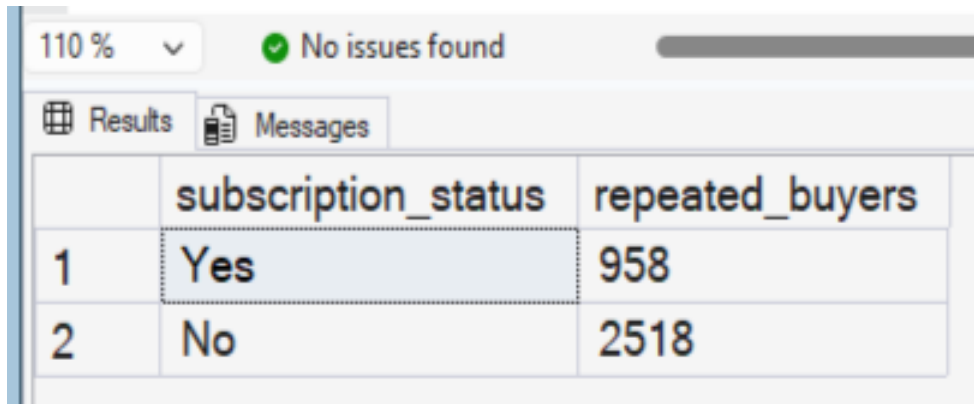
	customer_segment	Number of Customers
1	Returning	701
2	Loyal	3116
3	New	83

8. Top 3 Products per Category - Listed the most purchased products within each category.



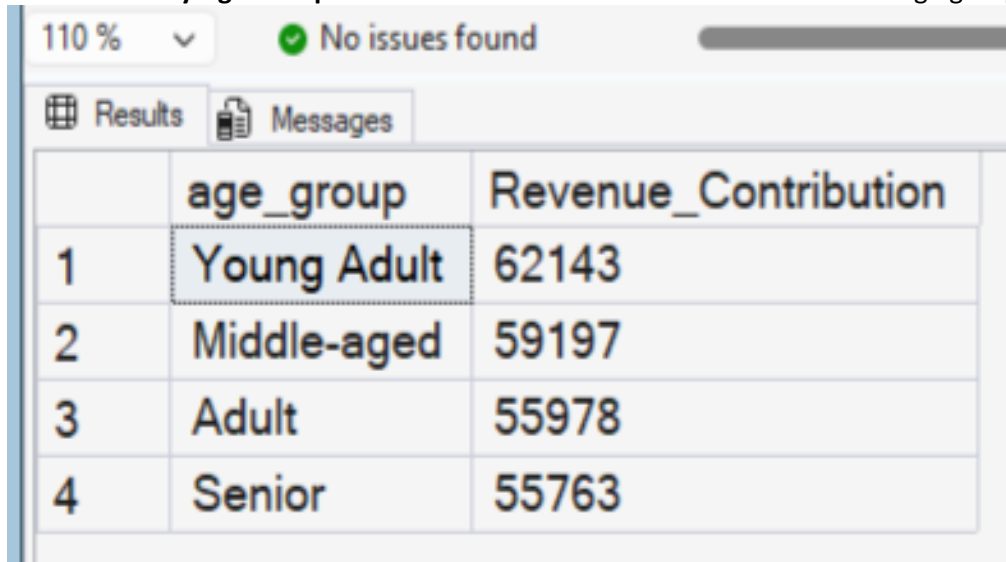
	item_rank	item_purchased	Total_Orders
1	1	Jewelry	171
2	2	Belt	161
3	3	Sunglasses	161
4	1	Blouse	171
5	2	Pants	171
6	3	Shirt	169
7	1	Sandals	160
8	2	Shoes	150
9	3	Speakers	145

9. Repeat Buyers & Subscriptions - Checked whether customers with >5 purchases are more likely to subscribe.



	subscription_status	repeated_buyers
1	Yes	958
2	No	2518

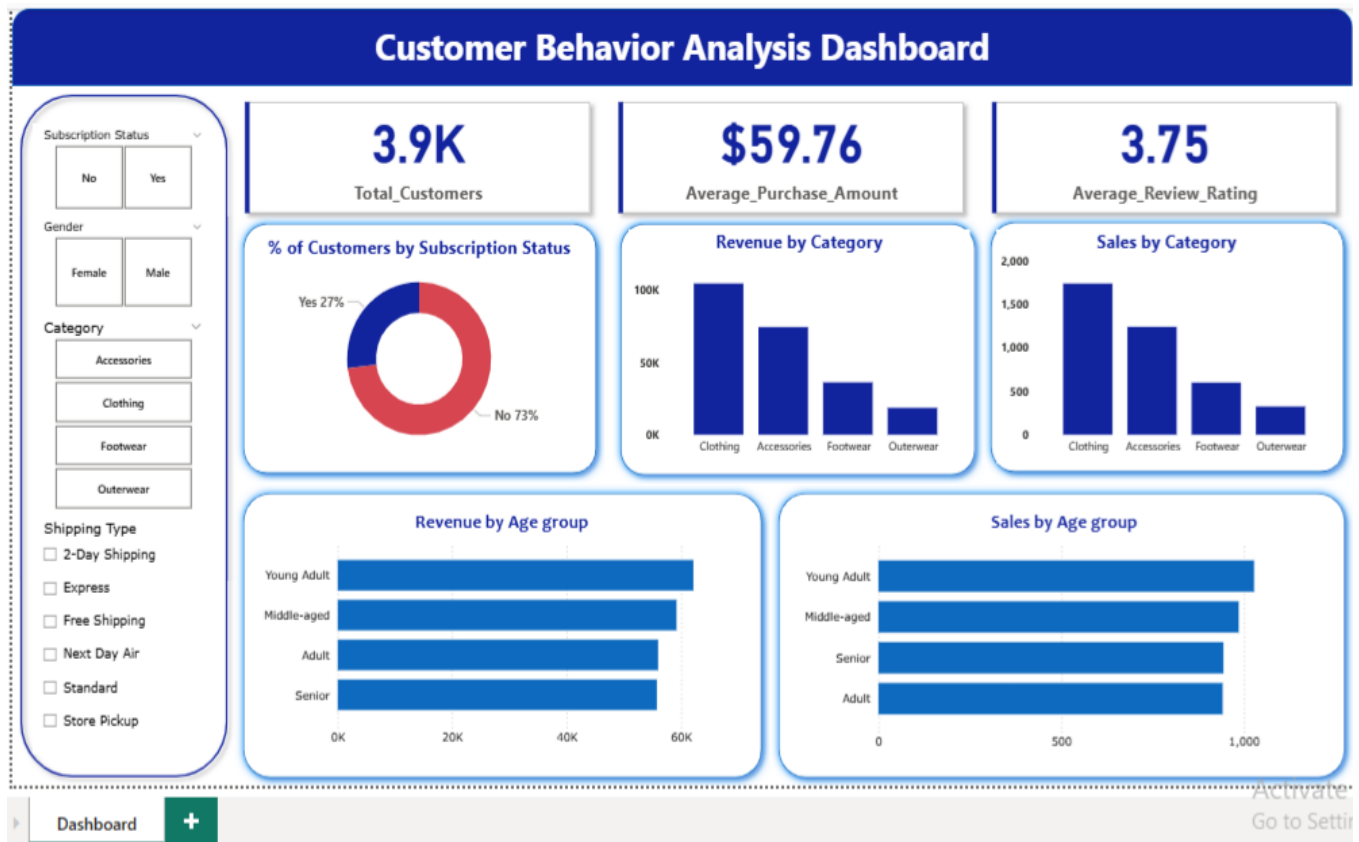
10. Revenue by Age Group - Calculated total revenue contribution of each age group.



	age_group	Revenue_Contribution
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Dashboard in Power BI:-

Finally, we built an interactive dashboard in Power BI to present insights visually.



Business Recommendations:-

- **Boost Subscriptions** - Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** - Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy** - Balance sales boosts with margin control.
- **Product Positioning** - Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** - Focus efforts on high-revenue age groups and express-shipping users.